A note on AI: We are aware that all of these tasks can be solved by AI within seconds. However, we encourage you to work on the tasks on your own first and, whenever you get stuck, ask us (or AI) for help. If you just let AI write the solutions, you won't deeply understand what you are doing – and we will notice this directly when we ask you to present your work.

Load the Handball, UEFA Euro football and Tennis data into your current R session.

1 Classification and Regression Trees (CART)

Work through the following steps for all three data sets, i.e., fit a regression tree to the handball data (`handball_wide`; response `scorediff`, predictors `bet.probs.diff` and `season`), a regression tree to the `Euro_train` data (response `Goals`, predictors `GroupStage`, `GDP`, `MarketValue`, `FifaRank`, `UefaPoints`, and `CLPlayers`), and a classification tree to the tennis data (response `win`, predictors `rank`, `points`, `elo`, and `age.30`).

1. Fit a regression or classification tree, respectively, to the data.
2. Load the `rpart.plot` package and draw the tree using the R function `rpart.plot()`.
3. Use the model to predict the goal difference (handball), the number of goals (football), or the winning probability (tennis) for each match using `predict()`.
4. Compute the mean absolute error (MAE) and root mean squared error (RMSE) of the predictions. For the tennis data, compute the ratio of correctly predicted matches instead.
5. Identify the matches with the largest overestimation and underestimation. For the tennis data, identify those predictions with the most extreme predicted winning probabilities. Discuss possible reasons based on the data.
6. Create a scatter plot of predicted vs. actual goal differences with the bisecting line as a reference to visually assess the model fit. For the UEFA Euro football data, use the number of goals instead. For the tennis data, create boxplots of the predicted winning probabilities separately for won and lost matches instead.
7. Take a closer look at the classification tree for the tennis data. How many different winning probabilities does it predict? Why do the boxplots from Step 6 look so different for won and lost matches? Why does the tree stop growing so early? (Hint: check `?rpart.control`, in particular the complexity parameter `cp`, and think about which error measure is used for classification trees.) Refit the tree with `cp = 0.001`. What changes?

2 Regression Trees and Overfitting

1. Fit a regression tree to the `Euro_train` data using the following predictors: `GroupStage`, `GDP`, `MarketValue`, `FifaRank`, `UefaPoints`, and `CLPlayers`. Use the default settings of `rpart()` (you can reuse the tree from Section 1).
2. Examine the importance of predictors in the regression tree. There are different types of feature importance measures. Find out which types exist, choose one of them and justify your choice. Which variables seem most influential for predicting goals?
3. Predict the number of goals for each match in the training data and the `Euro_test` data. Compute the MAE and RMSE.

4. Compare training and test errors. Discuss whether the tree is overfitting, underfitting, or reasonable.

5. Perform a grid search over the following tree hyperparameters:

   - `cp` $\in \{0, 0.001, 0.005, 0.01, 0.02, 0.05\}$
   - `minsplit` $\in \{2, 5, 10, 20, 40\}$
   - `maxdepth` $\in \{1, 2, 3, 5, 7, 10, 15\}$

   Fit a regression tree for each combination and evaluate MAE on training and test data.

6. Select the tree with the lowest test MAE. Report the corresponding parameters (`cp`, `minsplit`, `maxdepth`) and the corresponding training/test errors.

7. Visualize the best tree using `rpart.plot()` as well as the most overfitting tree (defined as the largest difference between training and test MAE). What do you notice?

8. Visualize how the training and test MAE change with increasing tree complexity, e.g., by plotting both errors against `maxdepth` (with `cp = 0` and `minsplit = 2`). What do you observe?

9. Repeat steps 5–8 for the handball and the tennis data. For the handball data, use the last season as test data and all other seasons as training data. For the tennis data, use the same split as in Sheet 1 (training data: 2011–2017, test data: 2018–2024) and the ratio of wrongly predicted matches instead of the MAE. Compare the results for the three data sets.